



Generative AI (GenAI) Notes

What is Generative AI?

Generative AI is a type of Artificial Intelligence that can create new content such as:

- Text
- Images
- Code
- Audio
- Video

Examples

- ChatGPT
- Gemini
- Claude
- GitHub Copilot

Traditional AI vs Generative AI

Traditional AI	Generative AI
Predicts	Creates
Classification	Content Generation
Fixed Output	Dynamic Output
Spam Detection	ChatGPT

How GenAI Works

```
graph TD; A[User Query] --> B[Prompt]; B --> C[LLM]; C --> D[Generated Response]
```

Example

User:

Explain Kubernetes

LLM:

Kubernetes is a container orchestration platform...

What is an LLM?

LLM stands for Large Language Model.

It is trained on huge amounts of data and can:

- Understand language
- Generate content
- Summarize information
- Generate code
- Answer questions

Popular LLMs

- GPT
- Gemini
- Claude
- Llama

Components of GenAI Architecture

```
graph TD; User --> ApplicationLayer[Application Layer]; ApplicationLayer --> OrchestrationLayer[Orchestration Layer]; OrchestrationLayer --> LLM[LLM]; LLM --> Response;
```

Enterprise GenAI Architecture

```
User
↓
Web App / Chatbot / Teams
↓
Authentication
↓
Orchestration Layer
↓
RAG Pipeline
↓
LLM
↓
Response
```

What is Prompt Engineering?

Prompt Engineering is the process of creating effective prompts to get better outputs from an LLM.

Basic Prompt

```
Explain Kubernetes
```

Better Prompt

```
Explain Kubernetes for a DevOps Engineer with examples.
```

What is RAG?

RAG stands for:

```
Retrieval Augmented Generation
```

RAG enables an LLM to use external documents while answering questions.

Why RAG?

Problem:

LLMs can hallucinate.

Solution:

Provide relevant documents before generating an answer.

Benefits:

- Better accuracy
- Reduced hallucinations
- Access to enterprise knowledge

RAG Architecture

Offline Phase

```
Documents
↓
Chunking
↓
Embedding Model
↓
Vector Database
```

Online Phase

```
User Query
↓
Embedding
↓
Similarity Search
↓
Relevant Chunks
↓
Prompt Builder
↓
LLM
```

↓
Response

What is Chunking?

Chunking is the process of splitting large documents into smaller pieces.

Example:

100 Page PDF
↓
Chunks
↓
Embeddings
↓
Vector Database

What are Embeddings?

Embeddings are numerical representations of text.

Example:

"Kubernetes"
↓
[0.12, 0.45, 0.89, ...]

Purpose:

- Similarity Search
 - Semantic Search
 - Recommendation Systems
-

What is a Vector Database?

A Vector Database stores embeddings and retrieves similar information.

Popular Vector Databases

- Pinecone
- ChromaDB
- Weaviate
- FAISS
- Azure AI Search

What is Agentic AI?

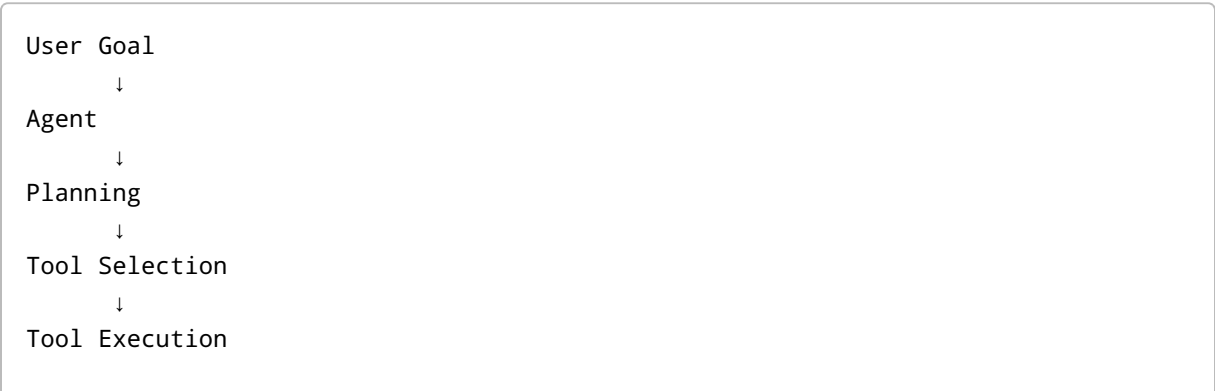
Agentic AI is an AI system that can:

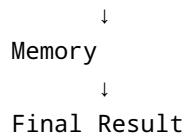
- Plan
- Reason
- Use Tools
- Execute Actions
- Maintain Memory
- Achieve Goals

GenAI vs Agentic AI

GenAI	Agentic AI
Generates Content	Completes Tasks
Single Response	Multi-Step Execution
Limited Autonomy	Higher Autonomy
No Tool Usage	Uses Tools

Agentic AI Architecture





Example Tools

- GitHub
- Terraform
- Azure CLI
- Kubernetes
- Databases
- REST APIs

GenAI Use Cases

DevOps

- YAML Generation
- Terraform Generation
- Shell Script Generation
- Documentation Creation

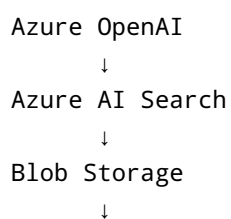
Software Development

- Code Generation
- Code Review
- Unit Test Creation

Business

- Chatbots
- Knowledge Search
- Document Summarization

Azure GenAI Stack



AKS



Azure Monitor

Important Interview Questions

What is GenAI?

AI that generates new content such as text, images, code, audio, and video.

What is RAG?

A technique that retrieves relevant documents before sending context to an LLM.

What is an Embedding?

A numerical representation of text used for semantic search.

What is a Vector Database?

A database optimized for storing and searching embeddings.

Difference Between GenAI and Agentic AI?

GenAI generates content while Agentic AI can plan, use tools, and execute tasks.

Learning Roadmap

Python



Pandas



NumPy



PyTorch



Transformers



Embeddings



Vector Database



RAG
↓
LangChain
↓
LangGraph
↓
Agentic AI

Key Takeaways

- GenAI creates content.
- LLM is the brain behind GenAI.
- RAG improves accuracy using external knowledge.
- Embeddings enable semantic search.
- Vector Databases store embeddings.
- Agentic AI can plan and execute tasks.
- Python is the foundation for learning GenAI.